IIIT KOTTAYAM | QUANTUM COMPUTING SEMESTER V | V2.0 VERIFIED

QForge: Dual Platform Operator Guide

Complete Step-by-Step Manual for `app_web.py` (Interactive Web Studio) & `app.py` (Desktop Control Workstation)

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> 00 // EXECUTIVE OVERVIEW

The **QForge Hybrid QCQ-CNN** platform provides two distinct, synchronized software interfaces designed for research execution and viva evaluation:

Interface	Script	Technology	Primary Use Case
Interactive Web Studio	`app_web.py`	Streamlit + PyTorch CPU	**Live Clinical Demonstration:** Drag-and-drop MRI inference, 4-channel quantum feature heatmaps, model comparisons, and interactive NISQ noise sliders.
Desktop Control Station	`app.py`	PySide6 (Qt) + Autograd	**Lab Workstation:** Automated 5-seed grid execution, training telemetry, loss convergence tracking, and publication figure generation.

> 01 // QUICKSTART LAUNCH COMMANDS

// OPTION A: LAUNCHING THE INTERACTIVE WEB STUDIO (RECOMMENDED FOR DEMOS)

Open a PowerShell terminal in the project directory and run:

```
.\.venv\Scripts\streamlit.exe run app_web.py
```

(Alternatively, double-click `run_web.bat` in File Explorer). Your browser will immediately open `http://localhost:8501`.

// OPTION B: LAUNCHING THE DESKTOP CONTROL WORKSTATION

In your PowerShell terminal, run:

```
.\.venv\Scripts\python.exe app.py
```

(Alternatively, double-click `run_app.bat` in File Explorer). A dark-mode desktop window will open with telemetry panels.

> 02 // HOW TO USE APP_WEB.PY (INTERACTIVE WEB STUDIO)

The Web Studio is designed for rapid interactive demonstration during project reviews and viva presentations. Follow these steps to demonstrate every capability:

01 **Load an MRI Scan for Diagnostic Inference**

In Tab 1 (`> 01 // LIVE INFERENCE & KERNELS`):

- Click `[ LOAD: GLIOMA SAMPLE ]` to load a malignant tumor with infiltrative margins.
- Click `[ LOAD: MENINGIOMA SAMPLE ]` to load a benign extra-axial lesion.
- Or drag-and-drop any clinical brain MRI slice (JPG / PNG) via the file uploader.

02 **Inspect 4-Channel Quantum Filter Feature Maps**

Observe the center panel: the 4-qubit quantum filter applies local unitary transformations on $2 imes 2$ pixel patches across the 28×28 input, outputting four $14 imes 14$ spatial expectation feature maps:

`Channel 0: <Z0> | Channel 1: <Z1> | Channel 2: <Z2> | Channel 3: <Z3>`

Explain to your faculty: *"These heatmaps visually reveal how the quantum Hilbert space maps tissue densities into orthogonal state projections."*

03 **Demonstrate the Correlated Pauli-Z Readout**

In the left sidebar under **Select Active Architecture**, switch between:

- **Model A (Frozen Random Baseline)**: Static uncalibrated quantum filter.
- **Model B (Trainable Filter)**: Gradient-optimized spatial quantum filter.
- **Model B + Correlated Readout (Full QForge Model)**: Demonstrates our multi-observable Pauli readout:
  `[<Z0>, <Z1>, <Z0*Z1>]`
  *Point to the joint term <Z0*Z1> at the bottom of Tab 1: it preserves 2-qubit entanglement correlations prior to decision classification!*

04 **Simulate Real-Time NISQ Hardware Noise (Chapter 4.6)**

In the left sidebar, check **Enable Simulated Quantum Noise**:

- Adjust the **Depolarizing Rate γ** slider from 0.00 to 0.05.
- Select **Measurement Shots** (128, 512, 1024, or Infinite).
- Observe in real time: Model A degrades rapidly under gate error, while Model B + Correlated Readout maintains robust diagnostic accuracy!

05 **Review Empirical Benchmark Statistics**

Click Tab 2 (`> 02 // BENCHMARK MATRIX & STATS`) to display the paired 5-seed benchmark table:

- **Model A:** 85.75% ± 1.66% Acc | 87.33% AUC | Drift: 0.00 rad
- **Model B:** 85.36% ± 1.14% Acc | 87.34% AUC | Drift: 1.83 rad

- **Model B + Correlated Readout: 87.25% ± 0.95% Acc | 88.90% AUC**

> 03 // HOW TO USE APP.PY (DESKTOP CONTROL WORKSTATION)

The desktop application (built with PySide6) serves as the primary research control console for executing experiments and inspecting hardware telemetry:

// DESKTOP NAVIGATION STRUCTURE

Use the left sidebar navigation bar to toggle between the four workstation modules:

- **01. CONTROL:** Configure hyperparameters (learning rate, filter depth, seeds, batch size). Contains the `[ RUN TRAINING ]` trigger.
- **02. EXECUTION:** Displays live terminal stdout/stderr stream from CPU statevector simulation with real-time epoch timers and gradient norms.
- **03. RESULTS:** Displays the complete 5-seed comparison matrix (Model A, Model B, Correlated Model in green) and hypothesis testing (p-values).
- **04. VISUALIZATIONS:** Interactive high-resolution figure browser. Click on: `[ FIG 1 · DYNAMICS ]`, `[ FIG 2 · DRIFT ]`, `[ FIG 3 · ROC ]`, `[ FIG 5 · FEATURE MAPS ]`, and our newly added **`[ FIG 6 · NISQ NOISE ]`**.

> 04 // FACULTY VIVA TALKING POINTS (KEY ANSWERS FOR HIGH MARKS)

Q1: Why is our accuracy ~87% when Long et al. (Nature 2025) claimed 95%?
Answer: "Long et al. evaluated raw Kaggle splits containing **110 duplicate images leaked across train and test sets**, creating synthetic memorization. We performed perceptual hash deduplication (`audit_and_deduplicate()`) to eliminate data leakage. On this rigorous leak-free benchmark, the true baseline is 85.8%, which our correlated model elevates to 87.3%."

Q2: What is the benefit of our Correlated Pauli Readout (<Z0>, <Z1>, <Z0*Z1>)?
Answer: "Prior literature measured only qubit 0 (<Z0>), mathematically tracing out qubit 1 and discarding non-local quantum entanglement correlations produced by the CNOT gates. By measuring the joint Pauli observable <Z0*Z1>, we preserve the full 2-qubit entangled state prior to the linear decision layer."

Q3: What do the NISQ hardware noise simulations show?
Answer: "Under physical depolarizing error rates typical of IBM superconducting devices (γ = 0.02 - 0.03), static random filters collapse by over 10.5%, whereas our adaptive correlated model drops by only 2.75%, retaining over 84.5% diagnostic accuracy under finite-shot sampling."

> 05 // SYSTEM VERIFICATION SUMMARY

```
[OK] All 17 unit tests passed in 6.03s (tests/test_qsim_and_models.py, tests/test_noise.py)
[OK] Feature branch 'feature/correlated-pauli-vqc' pushed to origin
[OK] Authored by: Yashwanth S <yashwanth.siiitk@gmail.com> (Team Lead — 2024BCS0264)
[OK] Both app_web.py and app.py operational and synchronized with results/summary.csv
```